

MUHAMMAD ALI HAIDAR

PHD CANDIDATE & RESEARCH ASSOCIATE

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PROFILE

PhD Candidate & Research Associate with 4+ years of experience in experimental life sciences, data analysis, and translational research. Skilled in experimental life sciences research across neuroscience, molecular diagnostics, and pre-clinical studies. Proficient in Python and R for reproducible data processing and analysis. Experienced in project management, delivering research and digital health work under tight deadlines. Designed behavioral assays and analysis pipelines, and supervised Master students and interns.

EDUCATION

Freie Universität Berlin, Berlin Dr. rer. nat. Biology Dissertation: Exploring the Relationship between Behavioral Individuality and Circuit Wiring in <i>D. melanogaster</i>	Oct 2026
American University of Beirut, Beirut Master of Science in Biochemistry GPA: 3.97/4 Thesis: Neuroprotective Effects of MitoQ in Open Head Traumatic Brain Injury	Sept 2021
University of Balamand, Beirut Bachelor of Science in Medical Laboratory Sciences GPA: 3.49/4	Sept 2016

EXPERIENCE

PhD Candidate & Research Associate, Freie Universitaet Berlin Berlin	Apr 2022 to Jul 2026
- Conducted rigorous experiments to explore neural circuit mechanisms influencing behavioral individuality in <i>D. melanogaster</i>, designing and implementing robust behavioral assays for both visual and non-visual responses. - Utilized confocal microscopy and quantitative morphometric analysis to map circuit wiring, linking structural variations to individual sleep and activity patterns. - Developed and maintained reproducible Python and R pipelines for the processing, analysis, and visualization of behavioral and morphological data. - Mentored Master students and interns in experimental design and data analysis, providing ongoing methodological guidance throughout their thesis projects. - Delivered comprehensive training in advanced neurobiological techniques as part of the developmental neuroscience course over three years.	
Technical Project Manager (Contract), LebMASH Beirut	Feb 2025 to Aug 2025
- Spearheaded the development of an AI-powered digital health platform, translating user needs into actionable requirements to enhance access to medical information for women and marginalized communities in Lebanon. - Orchestrated qualitative research across 7 distinct demographics, identifying user requirements to ensure the platform effectively addressed specific community health gaps. - Curated a culturally sensitive database of evidence-based health resources, optimizing digital content for the target audience to improve public engagement. - Directed project milestones within a 6-month funding window, ensuring timely completion of deliverables and stakeholder reporting.	
Research Assistant, American University of Beirut Beirut	Aug 2021 to Feb 2022
- Investigated pre-clinical drug candidates for open-head Traumatic Brain Injury models, assessing their efficacy in reducing oxidative stress and secondary inflammatory pathways. - Executed a series of behavioral assays, including memory, learning, and anxiety tests, to measure neurobehavioral recovery in subjects following treatment. - Conducted high-resolution imaging and molecular analysis, applying genetic and protein expression techniques to confirm the drug's effects on neuroinflammatory markers. - Authored a peer-reviewed publication detailing project findings, enhancing data visualization and technical analysis of the drug's mechanistic impact on secondary injury.	

- Adhered to strict animal surgical and care protocols, ensuring high-quality data collection in a standardized pre-clinical trial setting.

Molecular Technologist, Lebanese Ministry of Public Health

Oct 2020 to Sept 2021

Beirut

- Processed high-volume diagnostic samples at the national level for all international entries into Lebanon, achieving rapid turnaround times during the COVID-19 pandemic.
- Executed comprehensive molecular diagnostic protocols, including automated and manual RNA extraction and RT-qPCR for viral detection.
- Maintained rigorous biosafety and quality control standards in a high-throughput laboratory environment, ensuring 100% diagnostic accuracy and data integrity.
- Streamlined sample intake and reporting workflows in collaboration with a multidisciplinary team to effectively manage increased national testing demands.

SKILLS

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| • Experimental Life Sciences | • Data Analysis (Statistical Modeling, R, Python) | • Laboratory-Based Discovery | • Computational Approaches |
| • Communication | • Mentoring | • Collaboration | • Interdisciplinary Teamwork |
| • Public Engagement | • Behavioral and Molecular Techniques | • In-vivo skills (Murine and Insects) | |

LANGUAGES

English, Arabic, German

REFERENCES
